

TRAN DAI NHAN

AI Engineer | Applied NLP, LLM Systems and Backend Services

Ha Noi Capital, Vietnam | (+84) 355 126 752 | nhantd.dev@gmail.com

linkedin.com/in/nhantran8104 | github.com/hugebenevolence | hugebenevolence.github.io | ORCID
0009-0009-6020-3591

EDUCATION

FPT University

Oct 2022 - Jan 2026

Bachelor of Engineering, Software Engineering. Major in Software Development

GPA 3.3 / 4.0

- Capstone: S.M.I.L.E, a multi clinic dental practice management platform. Owned appointment booking, payments and identity verification across a microservice backend.
- Coursework: distributed systems, database systems, software architecture, machine learning, computer vision, data structures and algorithms.
- Produced two accepted conference papers, founded an AI startup and led the university IT club alongside full time study.

TECHNICAL SKILLS

Languages	Python, TypeScript, Java, C#, SQL, Bash
AI and ML	PyTorch, Transformers, vLLM, Unsloth, LLaMA-Factory, MS-SWIFT, LangGraph, LangChain, Qdrant, Hugging Face
AI Practice	LLM fine tuning (SFT, GRPO), knowledge distillation, RAG and retrieval evaluation, agentic workflows, multimodal VQA, vision language pretraining, inference cost optimisation, benchmark design
Backend	FastAPI, NestJS, Spring Boot, .NET, REST API design, microservices, concurrency control, PostgreSQL, Redis, TypeORM
Cloud and DevOps	AWS (EC2, S3, IAM), Docker, Docker Compose, GitHub Actions CI/CD, Nginx, Cloudflare, Vercel, Linux
Practices	Agile and Scrum, code review, automated testing (Jest, pytest, Vitest)

EXPERIENCE

Vingroup - AI Talent Training Program

Sep 2026 - Present

AI Trainee, Team Lead

Hanoi Capital

- Led a team building a multi-agent system to validate and reproduce baseline papers, parsing paper claims, repository artifacts and GitHub issue threads into a single reconciled completeness checklist.
- Designed the four-stage pipeline (Paper Agent, Repo Agent, Issue Agent, Reconciler) and scoped agent authority to collection and diagnosis only, keeping code generation and final judgment with the human reviewer.

AI VIET NAM

Oct 2025 - Present

AI Researcher and Teaching Assistant

Ho Chi Minh City

- Run the experimentation workflow for Vietnamese VQA and NLP research: dataset curation, training runs, evaluation harnesses and result reporting for the team.
- Mentor learners through hands on labs in NLP and multimodal AI, translating current research into exercises engineers can reproduce.
- Author lecture slides, lab notebooks and coding exercises for AI training cohorts, and review assessment material for accuracy.

FPT Software HCM

Jan 2025 - Jul 2025

Web Developer Intern, Backend

Ho Chi Minh City

- Built backend modules for a course management platform in Java and Spring Boot over PostgreSQL, adding Redis caching and Docker containerisation to improve response times and deployment consistency across Agile sprints.

RESEARCH AND PUBLICATIONS

Difficulty-Aware Adaptive Reasoning for Vietnamese VQA with GPT-OSS

Accepted, AICI 2026

First author. Springer book chapter. PyTorch, Transformers, vLLM, Unsloth, Gemini 2.5

Code

- Designed a modular Vietnamese VQA framework covering dense image captioning, multi LLM inference across GPT-OSS, Qwen3 and DeepSeek, batch processing and automated evaluation.
- Built a difficulty aware router that scales inference compute per question across low, high and adaptive modes, cutting end to end inference from 3,317s to 815s, a 75 percent reduction against the high compute baseline.
- Reached BLEU@4 0.142, ROUGE-L 0.654 and METEOR 0.421 on ViVQA-X at a quarter of the baseline compute budget.

- Built the distillation pipeline curating low, medium and high budget reasoning traces from a GPT-OSS teacher for a Llama 3.2 3B student.
- Added final answer verification and length based filtering to strip noisy and over long chain of thought supervision, prioritising sample quality over volume.
- Ran SFT and GRPO workflows reaching GSM8K 0-shot 0.8006 and MATH500 0-shot 0.4760.

SELECTED PROJECTS

S.M.I.L.E, Dental Practice Management Platform | Capstone

2026

Booking, Payment and Identity Owner: NestJS, Bun, TypeORM, PostgreSQL, Redis, LangGraph, Qdrant, Python

Code

- Built the appointment booking APIs for a multi clinic network, using database level concurrency control on doctor time slots so two patients cannot claim the same slot under simultaneous requests, with realtime availability updates in the UI.
- Owned the payment service end to end inside a 4 service, 5 database microservice platform, covering VNPay initiation by QR and redirect, callback handling, refund approval workflow and payment history over its own PostgreSQL bounded context.
- Eliminated double charge risk under client retries with Redis backed idempotency keys, and hardened the VNPay callback with signature verification and replay protection.
- Shipped an agentic booking assistant on LangGraph and FastAPI, with Qdrant vector retrieval for clinic policy questions, routed through the API gateway and wired into the booking flow.
- Built the KYC API for Vietnamese citizen ID cards as a standalone Python OCR service extracting and validating identity fields, wired into the IAM service to gate booking eligibility.

Vietnamese Vision Language Model Pretraining from Scratch

2026

Team project. SigLIP2, MLP projector, Llama 3.2 1B Instruct, PyTorch, Accelerate, Hugging Face

Code

- Implemented a two stage LLaVA style curriculum: stage 1 freezes the SigLIP2 vision encoder and the Llama decoder and trains only the MLP projector for modality alignment, stage 2 unfreezes the decoder for Vietnamese visual instruction tuning with scraping, generated data from OpenAI Batching.
- Ran distributed training with Accelerate using checkpoint and resume handling so long runs survive interruption, then closed the loop with benchmark evaluation and working demos for captioning and visual instruction chat.

Vietnamese GPT-2 Pretraining from Scratch

2026

Team project. PyTorch, Transformers, FastAPI, Docker, Next.js

Code

- Trained a Vietnamese subword tokenizer from scratch rather than reusing an English vocabulary, cutting token fragmentation on Vietnamese text and improving the effective context budget.
- Curated and deduplicated a mixed news and Wikipedia corpus into roughly 2.64B training tokens across pretraining and continued pretraining phases.
- Adapted the model to a narrow five word quatrain corpus as a controlled study of how far continued pretraining shifts generation style, then packaged it for serving as a Docker based trainer with a FastAPI backend and a Next.js chat interface.

PawCal, AI Veterinary Automation Platform

2025 - Present

Founder and Technical Lead. Next.js, Supabase, n8n, Zalo OA, Vercel, Docker

- Built a chatbot agent on Zalo OA handling appointment scheduling, automated reminders and customer enquiries for veterinary clinics.
- Secured roughly USD 2,000 in early support and validated the product with beta clinics.

LEADERSHIP, AWARDS AND CERTIFICATIONS

President, FCODER, FPT University Information Technology Club

Oct 2023 - Oct 2025

- Directed club operations, coding competitions and technical training; ranked 1st among academic clubs at FPT University in 3 of 4 terms.
- Proposed and led project initiatives presented to the leadership board of FPT University's academic clubs union.

Research Leader, FPT Research Festival

2025

- Led a student research team through problem framing, experiment planning and paper writing, delivering an image labelling pipeline for breast cancer data and an optimised clinical prediction workflow across 2 accepted papers.

Awards | First Prize (team), Coca-Cola environmental protection solution challenge. Top 2, FPT Can Tho AI Student Olympiad 2026. Top 10, FPT Hackathon (AI support platform for children with autism). Provincial Excellent Student Award, Chemistry.

Certifications and Languages | Software Development Lifecycle (Coursera). MOS Word 950, Excel 975. Vietnamese native, English professional working proficiency.